

Pancreatic adenocarcinoma

>90% of pancreatic cancers (PNET <5%) · median age 71 · 60–70% arise in the head · 5-yr relative survival — localized **44%**, regional **16%**, distant **3%**; **13%** all stages, **8%** for adenocarcinoma

Stanford Oncology Handbook 2025, Pancreatic Cancer (Johnson T, Lee W) · NCCN v2.2025 · items marked † are outside the handbook chapter

WORKUP — ALL PATIENTS

- **Pancreatic-protocol CT** chest/abdomen/pelvis — required for staging and resectability
- **EUS with biopsy** if clinically indicated
- MRI for indeterminate liver lesions or local extent
- Labs: CBC, CMP, baseline **CA 19-9**
- **Germline testing in ALL patients** regardless of family history — ~10% harbor a mutation
- **Tumor genomic profiling** in all metastatic patients
- Manage at a **high-volume center** with multidisciplinary input

PRESENTATION

- Early: **weight loss, nausea, bloating, abdominal pain**
- “Painless jaundice” is common but **by no means universal**
- † New-onset diabetes in an older adult; migratory thrombophlebitis
- † **CA 19-9 pitfalls**: falsely elevated by biliary obstruction — recheck after decompression; **5–10% are Lewis-antigen negative** and never secrete it

STAGING — AJCC 8TH

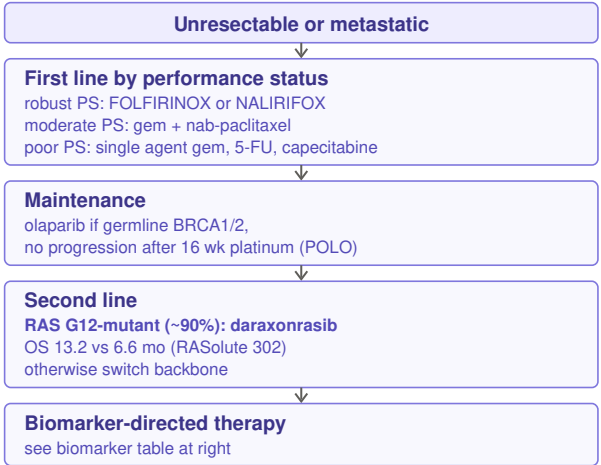
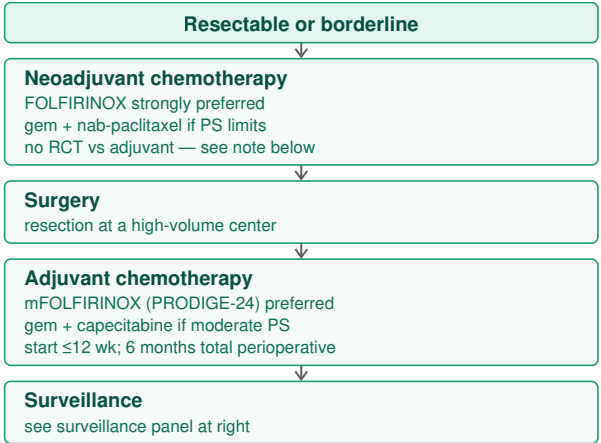
T1 / T2 / T3	≤2 cm / >2–4 cm / >4 cm
T4	celiac axis, SMA, or common hepatic artery — regardless of size
N1 / N2	1–3 nodes / ≥4 nodes
Stage III / IV	T4 or N2 / M1

In practice resectability — **not TNM** — **drives management**: resectable · borderline · unresectable localized · metastatic. The call belongs to a hepatobiliary surgeon experienced with the Whipple.

KEY TRIALS

PRODIGE-24	adjuvant mFOLFIRINOX — OS 54.4 vs 35.0 mo (NEJM 2018)
ESPAC-4	adjuvant gem + capecitabine (Lancet 2017)
PRODIGE-4	FOLFIRINOX in metastatic disease (NEJM 2011)
MPACT	gem + nab-paclitaxel (NEJM 2013)
POLO	olaparib maintenance, germline BRCA (NEJM 2019)
KEYNOTE-158	pembrolizumab in MSI-H (JCO 2020)
RASolute 302	daraxonrasib, 2L metastatic — median OS 13.2 vs 6.6 mo in RAS G12-mutant (NEJM 2026)

TREATMENT PATHWAY



Neoadjuvant therapy has no RCT support. No prospective trial shows it beats adjuvant therapy; use is extrapolated from PRODIGE-24 and justified because it identifies bad biology before a morbid operation and gives early systemic control.

Two comparisons never made head-to-head: adjuvant mFOLFIRINOX vs gem/capecitabine, and first-line metastatic mFOLFIRINOX vs gem/nab-paclitaxel — **choose by performance status.**

REGIMENS & DOSING

Regimen	Dosing	Cycle
FOLFIRINOX	oxaliplatin 85 · irinotecan 150 · 5-FU 1200/day CIV d1–2	q14d
NALIRIFOX	5-FU 1200 d1–2 · liposomal irinotecan 50 · oxaliplatin 60	q14d
Gem + nab-paclitaxel	gemcitabine 1000 + nab-paclitaxel 125, d1, 8, 15	q28d
Gem + capecitabine	gemcitabine 1000 d1, 8, 15 + capecitabine 850 BID d1–21	q28d
Gemcitabine	1000 mg/m ² d1, 8, 15	q28d
Capecitabine	1250 mg/m ² PO BID d1–14	q21d
5-FU	1200 mg/m ² /day CIV d1–2	q14d

All doses mg/m² unless stated. PRODIGE-24 used irinotecan 180 mg/m², reduced to **150 mg/m²** — preplanned subgroup analysis showed **equal efficacy**.

BIOMARKER → THERAPY

Alteration	Therapy
RAS G12 mutation (~90% of PDAC)	Daraxonrasib — oral multi-selective RAS(ON) inhibitor; metastatic disease after ≥1 prior line, or not a candidate for multiagent therapy
Germline BRCA1/2	Olaparib 300 mg PO BID — maintenance after ≥16 wk platinum (POLO)
MSI-H / dMMR / TMB >10	Pembrolizumab 200 mg q3wk or 400 mg q6wk
NRG1 fusion	Zenocutuzumab (Bizengri) 750 mg IV q2wk
BRAF V600E	Dabrafenib 150 mg BID + trametinib 2 mg daily
NTRK fusion	Entrectinib 600 daily · larotrectinib 100 BID · repotrectinib 160
RET fusion	Selpercatinib 120 mg BID (<50 kg) or 160 mg BID (≥50 kg)

Actionable alterations were historically rare — **daraxonrasib changes that**, since ~90% of PDAC is KRAS-mutant. Still profile every metastatic patient for the remainder.

SURVEILLANCE AFTER RESECTION

- H&P **q3–6 months × 2 years**, then q6–12 months
- CT chest/abdomen/pelvis with contrast q3–6 months × 2 years, then q6–12 months
- **CA 19-9** q3–6 months × 2 years
- **Signatera** or equivalent ctDNA q3–6 months if indicated

SUPPORTIVE CARE

- Nutrition referral · palliative care for all metastatic patients · hospice as appropriate
- † **Pancreatic enzyme replacement (PERT)** — chronically underprescribed; exocrine insufficiency drives weight loss and treatment intolerance
- † **VTE prophylaxis** — among the highest-risk malignancies for thrombosis
- † Biliary stenting for obstruction · celiac plexus neurolysis for refractory pain